

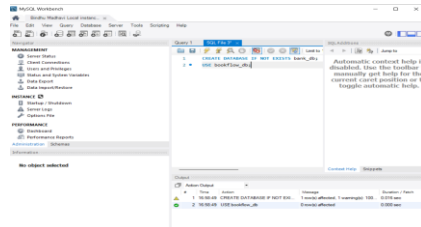
WEEK-2

Name: P. Bindhu Madhavi

Roll Number: 2520030077

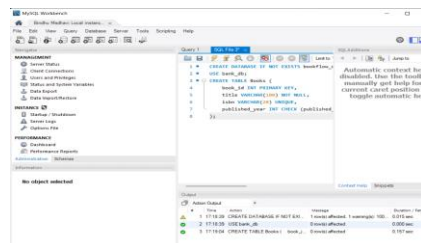
Step 1 – Create Database

Creates a new database to hold all the tables required for the Library Management System.



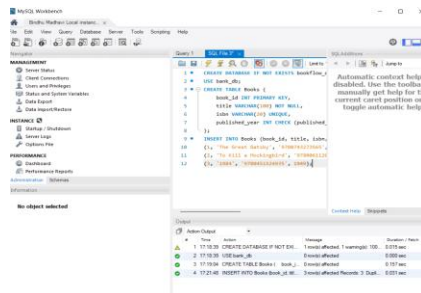
Step2-Create Books Table (DDL + Constraints)

Defines the Books table with a primary key, NOT NULL, and UNIQUE constraints to maintain data integrity.



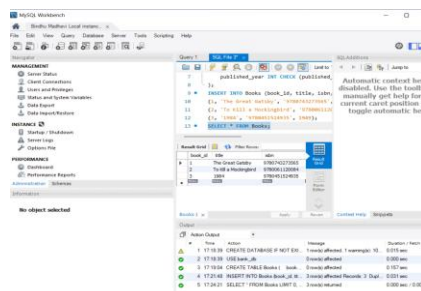
Step 3 - Insert Books

Inserts sample book records into the Books table using INSERT statements.



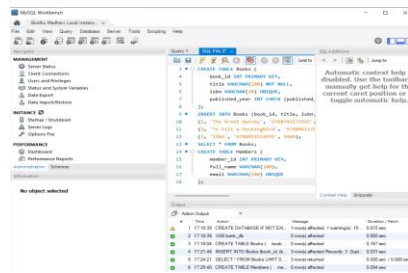
Step 4 - Display Books

Retrieves and displays all records currently stored in the Books table.



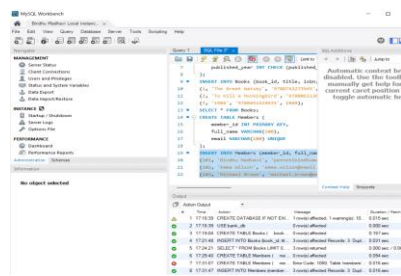
Step 5 - Create Members Table

Creates the Members table to store the personal and contact details of library members.



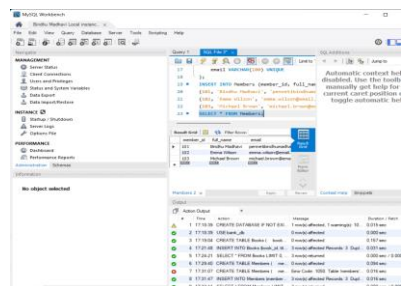
Step 6 - Insert Members

Adds sample member records into the Members table.



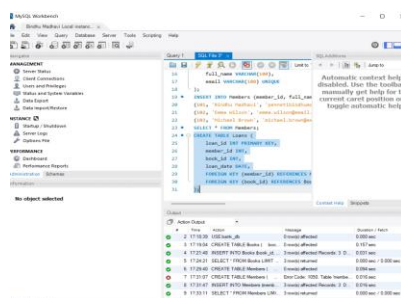
Step 7 - Display Members

Displays all member records to verify that the data was inserted correctly.



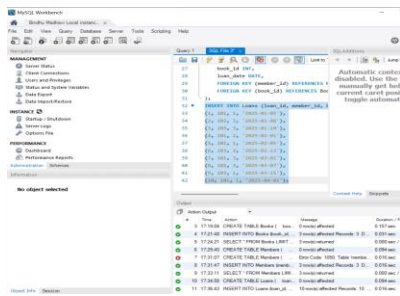
Step 8 - Create Loans Table

Creates the Loans table with foreign keys referencing the Books and Members tables.



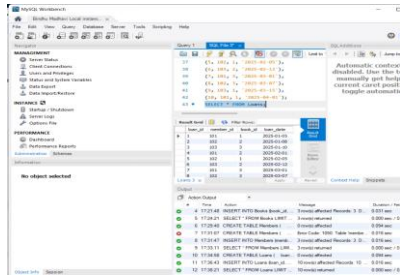
Step 9 - Insert Loans

Inserts sample loan transactions that link members to the books they have borrowed.



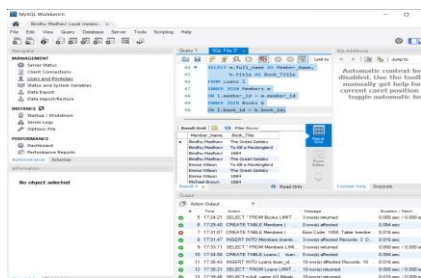
Step 10 - Display Loans

Displays all loan records to confirm the borrowing transactions were recorded.



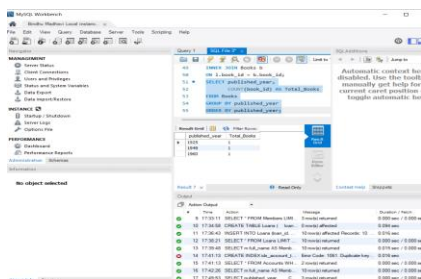
Step 11 - JOIN Query

Joins the Books, Members, and Loans tables to show which member borrowed which book.



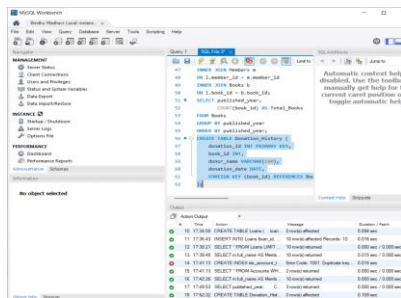
Step 12 – Group By Query

Groups loan records by member to count how many books each member has borrowed.



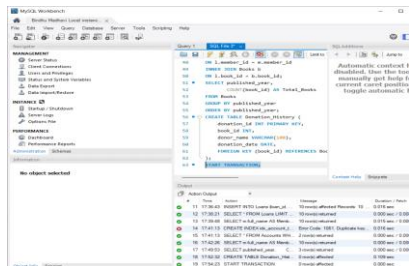
Step 13 - Create Donation History Table

Creates a table to track book donations made by library members.



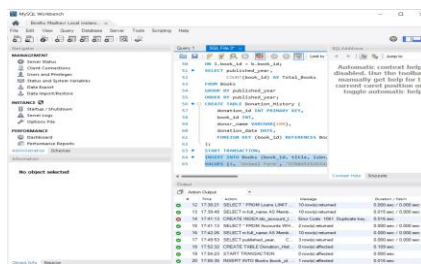
Step 14 - Start Transaction

Begins a transaction so that multiple related operations can be executed and committed together.



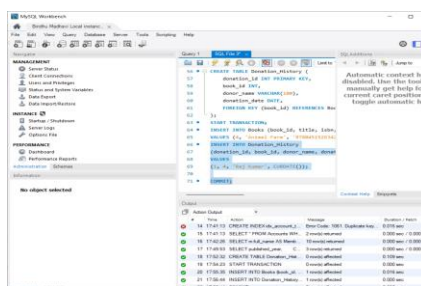
Step 15 - Insert New Book

Inserts a new book record as part of the ongoing transaction.



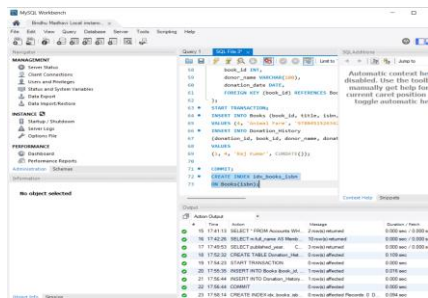
Step 16 - Insert Donation History and Commit

Records a donation entry and commits the transaction to permanently save the changes.



Step 17 - Create Index

Creates an index on a column to speed up future search queries.



Step 18 - Search Using Index

Executes a search query that uses the newly created index for faster retrieval.

